

Homework 5

March 12 - 19

Reading

1. Speech and Language Processing: I added this book to the GitHub. Start reading Chapter 3: *N*-gram Language Models. This section will put MLE in the context of natural language processing, so it should feel pretty natural after our linear regression discussion.

Problems

Problem 1. Use induction to prove the *chain rule of probability*: For n random variables $X_1, \dots, X_n$,

$$P(X_1, \dots, X_n) = \prod_{k=1}^n P(X_k \mid X_{1:k-1}).$$

(Here, $X_{1:k-1}$ refers to the joint distribution $X_1, \dots, X_{k-1}$).